REMINDER

ROLE OF EUROPEAN MOBILITY AND ITS IMPACTS
IN NARRATIVES, DEBATES AND EU REFORMS

REMINDER: Short Term Media Analysis on Migration 2017-2018 Documentation

Authors: Tobias Heidenreich[1], Fabienne Lind[1], Jakob-Moritz Eberl[1], Sebastian Galyga[1], Rachel Edie[1], Beatriz Herrero-Jiménez[2], Eva Luisa Gómez Montero[2], Rosa Berganza[2] & Hajo G. Boomgaarden[1]

[1] University of Vienna
[2] Universidad Rey Juan Carlos

Published: January 2020
Documentation prepared as part of the REMINDER project
www.reminder-project.eu

doi: 10.11587/T86DVG
AUSSDA Dataverse

Correspondence address:

Hajo Boomgaarden, University of Vienna, Department of Communication
Rathausstraße 19, 1010 Vienna, Austria, email: hajo.boomgaarden@univie.ac.at

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 727072

Contents

INTRODUCTION 2

CONDITIONS OF USE 3
Citation requirement 3
Confidentiality 3
Notification requirement 3

STUDY DESCRIPTION 4

DATA SET DESCRIPTION 5

MEDIA SOURCES 6

ARTICLE SELECTION 7
Migration - concept definition 7
Search string creation and validation 7
Article gathering and parsing 9

DATA CLEANING AND PREPARATION 10
Deduplication 10
Stratified sampling and article translation 10

METADATA 14

KEY VARIABLES 15
Intra-EU mobility 15
Concept Definition 15
The Dictionary 16
Sentiment towards migration 16
Migration-related frames 17
Concept Definition 17
Dictionary creation and data annotation 21
Validation procedure 22
Migration-frame related sentiment 23
Geographic regions 23

VARIABLE OVERVIEW 25

REFERENCES 25

REMINDER

Introduction

This Documentation accompanies the data of the REMINDER automated content analysis of migration-related media coverage published in seven European countries (namely, Spain, United Kingdom [UK], Germany, Sweden, Poland, Hungary, and Romania) between December 2, 2017, and November 6, 2018.

The Documentation is structured in nine key sections. First, the conditions of use of the data are outlined. The second section, the study description, provides an overview of the study, including information on the relevant files and topics. The subsequent section presents a short description of the data set. The fourth section gives an overview of the media sources. The fifth section details our approach to data collection. This section is followed by a description of our procedures in terms of data cleaning and data preparation. The seventh section contains information on first variables in the data set; so-called metadata. The eighth and largest section describes and defines key concepts/variables in the data set as well as validation procedures where necessary. The last section provides an overview of the variables in the data set.

The used dictionaries are available upon request.

Funding

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement no 727072.

Principal Investigator

Prof. Hajo G. Boomgaarden

Data file name

10086_mr_en_v1_0.pdf (Documentation and Method report)
10086_da_en_v1_0.dta (Data file, STATA format, Version 1.0)
10086_da_en_v1_0.zsav (Data file, compressed SPSS format, Version 1.0)

Topics/Keywords

newspaper coverage, online news coverage, automated text analysis, migration, EU mobility, sentiment, framing

Conditions of use

Citation requirement

Data users are kindly asked to provide references to the data and the accompanying release documents to all forms of publications (including working papers, theses, etc.) based on REMINDER data.

Please cite as:

Heidenreich, T., Lind, F., Eberl, J.-M., Galyga, S., Edie, R., Herrero-Jiménez, B., Gómez Montero, E.L., Berganza, R. & Boomgaarden, H.G. (2020). *REMINDER: Short Term Media Analysis on Migration 2017-2018 (OA edition).* (10086_da_en_v0_1.dta; Version 1.0), [Dataset and documentation]. AUSSDA Dataverse. doi: 10.11587/T86DVG

Confidentiality

All reasonable precautions have been taken by the authors to ensure the accuracy of the data set.

REMINDER, the authors and the funding institution bear no responsibility for the use of the data, or for interpretations or inferences based on their use, neither do they accept liability for indirect, consequential or incidental damages or losses arising from the use of the data collection.

Notification requirement

In order to facilitate exchanges within the scientific community and to provide funding agencies with essential information about the use of archival resources, users of the REMINDER data are requested to notify the REMINDER team of all forms of publications referring to REMINDER data (email: hajo.boomgaarden@univie.ac.at).

Study description

In order to put the data set in a broader context, we will briefly describe the project behind it:

REMINDER (**www.reminder-project.eu**) brought a multidisciplinary approach to understanding free movement in Europe, combining expertise from different fields, including development, economics, linguistics, media studies, political science and public policy.

In particular, the project had two components:

- The nature and impacts of intra-European-Union (intra-EU) mobility: REMINDER assessed the drivers of intra-EU mobility and how EU mobility has affected economies and labour markets in both origin and destination countries. The project also explored the impact of the movement of EU workers for public services and welfare systems designed in different ways.
- Political and media narratives about intra-EU mobility: REMINDER examined how different media portray intra-EU mobility and what factors drive individuals' views on free movement, including narratives used in party political communications and traditional and social media. The project also examined how media narratives on intra-EU mobility differ across Member States.

Using the information from these two components, the project then assessed policy options to address real and perceived impacts and pressures that free movement and other kinds of migration may generate.

These project objectives were addressed by different work packages. This Documentation and the data set associated with it are related to Work Package 8 (WP8). WP8 served as a cornerstone for the REMINDER project in that it established, in a systematic manner, the types of EU migration and mobility discourses in mass-mediated communication in Spain, the UK, Germany, Sweden, Poland, Hungary, and Romania. It did so by relying on large-scale text analysis techniques. The materials used were machine-readable texts from the various domains, including digitally archived online and print newspaper articles. By these means, WP8 enabled an in-depth assessment of how intra- and non-EU mobility are discussed and framed in different countries and at different points in time, which will contribute to an improved understanding of the interlinkages between mediated discourses, politics and public perceptions.

Linkage with REMINDER Survey Data

Please note that the media data from the "REMINDER: Short Term Media Analysis" can be linked to the "REMINDER: Online-Panel Study on Migration and Mobility Attitudes 2017-2018" dataset[1]. For an overview and discussion on how to link media and survey data, see de Vreese and colleagues (2017).

[1] See doi:10.11587/LBSMPQ, AUSSDA Dataverse.

Data set description

The data set is based on a multilingual text corpus. This text corpus comprises migration-related news articles (print and online) published between December 2, 2017 and November 6, 2018, in seven different European countries: Spain, UK, Germany, Sweden, Poland, Hungary, and Romania. Accordingly, the original articles are in Spanish, English, German, Swedish, Polish, Hungarian, and Romanian.

The dataset contains annotations for a stratified sample of media data. The data gathering, annotation, and cleaning was conducted between January 1, 2017, and December 31, 2019, using automated content analysis methodology.

In order to work with the multilingual material, seven native speakers (one per language) have been consulted at different stages of the project. We are grateful to Rachel Edie, Zehra Brakmic, Veronika Wenninger, Orsolya Fehér, Rubén Tamboleo García, Ewelina Marszowska, and Volina Șerban for their contribution throughout this process. All native speakers have excellent knowledge of the individual national contexts and a graduate degree in communication, translation, or linguistics. They were involved in the creation and validation of automated coding instruments.

Media sources

Table 1 gives an overview of the media sources that are included in the data set. The media sources were selected on the basis of reach, genre, and data availability, with the aim of acquiring a broad spectrum of each country's media environment. Please note, among the incorporated Swedish media outlets are three "alternative" media, labelled as "Online / Alternative" (`source_type`).

Table 1 Media sources in the data set

Country	Source type	Source
Spain	Print	El Mundo, El Pais
	Online	20minutos.es, abc.es, eldiario.es, elmundo.es, elpais.com, larazon.es, lavanguardia.com
UK	Print	Daily Mail, Daily Mirror, Metro, The Daily Telegraph, The Guardian,
	Online	bbc.com/news/uk, dailymail.co.uk, express.co.uk, mirror.co.uk, theguardian.com/uk, telegraph.co.uk, thesun.co.uk, thetimes.co.uk
Germany	Print	Bild, Die Tageszeitung (taz), Die Welt, Frankfurter Allgemeine Zeitung, Frankfurter Rundschau, Süddeutsche Zeitung
	Online	bild.de, faz.net, spiegel.de, sueddeutsche.de, taz.de, welt.de, zeit.de
Sweden	Print	Aftonbladet, Expressen, Svenska Dagbladet
	Online	aftonbladet.se, dn.se, expressen.se, metro.se, nyheter24.se, svd.se
	Online / Alternative	etc.se, friatider.se, samnytt.se
Poland	Print	Dziennik Gazeta Prawna, Gazeta Wyborcza, Rzeczpospolita
	Online	fakt.pl, gazeta.pl, onet.pl, rp.pl, se.pl, wp.pl, wyborcza.pl
Hungary	Print	Magyar Hirlap, Magyar Idök, Nepszava
	Online	24.hu, blikk.hu, borsonline.hu, index.hu, magyarhirlap.hu, mno.hu, napi.hu, nepszava.hu, ripost.hu
Romania	Print	Evenimentul Zilei, Jurnalul National, Romania Libera, Ziarul Financiar
	Online	adevarul.ro, click.ro, evz.ro, jurnalul.ro, libertatea.ro, romanialibera.ro, zf.ro, ziare.com

Article selection

Migration - concept definition

The explicit goal of our article selection process was to identify all news articles that contain information on migration or migrants.

“Migrant/s”, refers to people that explicitly changed, change or will/might change their place of residence from one country to another (i.e. refugees, asylum seekers, female/male migrants, immigrants, emigrants, citizens with migration background) – either wilfully or not. The term “migration” is understood as a generic term and thus stands equally for migration, emigration and immigration. An article “contains information” on this key concept as soon as something relating to migration/migrants is mentioned in the article. Migration does not have to be the main topic or a major topic of the article.

The decision whether an article contains information on migration or migrants is not affected by
a) the location: References to migrants or migration can be made, for example, about migrants being still in the sending country, already in the receiving country, or in the process of migration.
b) the time: References to migrants or migration can be made about something that has happened in the past, is happening right now or may happen in the future.
c) the (grammatical) modality: References can further be made in any modality, for example, realis mood (e.g. statement of fact, something is actually happening) or irrelative mood (e.g. hypothetical, conditional, wish, desire) of reported events.

To give a few examples for the scope of involved aspects, migration can be addressed in terms of its (legal, social, economic, political, environmental etc.) causes, developments, processes, phenomena and consequences. It can be referred to by mentioning involved individuals/groups or dependents in relation to the migration topic (migrants, people in sending, transit, and receiving countries, politicians, officers, social workers, etc.), in the discussion of policies (e.g. freedom of movement), or in the context of a fictional or non-fictional context.

Search string creation and validation

We used Boolean search strings for each country (and thus language) to select news articles that deal with migration. The Boolean search strings were designed to represent the above-introduced concept definition and thus include keywords that relate to the topics of immigration, emigration, migration in general, and freedom of movement.

The search strings were constructed with the support of native speakers. The goal was to make them as comparable as possible across language and country while considering the country- and language-specific particularities. The search strings are thus not fully identical for every language. Some keywords in one language may have several synonyms in another language (e.g. the English verb ‘to immigrate’ is captured with three synonymously used equivalents in the German: ‘immigrieren’, ‘einwandern’, ‘zuwandern’). In addition, for some

countries, context-specific terms were added (e.g. the Spanish keyword ‘patera’ translates to a small boat, frequently used by irregular migrants to cross the Mediterranean Sea).

Search strings were validated on language-specific text corpora consisting of articles from different national news outlets. These article subsets were manually annotated by native speakers. They were advised to read the articles of their subset carefully and to decide whether they mention migration-related aspects that meet the above-presented concept definition. The search strings were applied to the same set of articles. The classification decision of manual coders and search string were then compared.

With an average recall of 0.81 and average precision score of 0.85, the final search strings represent an appropriate tool to recognise migration-related texts (Lind et al., 2019). The detailed recall and precision scores are listed in Table 2. The final Boolean search strings are displayed in Table 3.

Table 2 Recall and precision for the Boolean search strings to recognise migration-related articles

Language	Number of manually annotated articles	Number of articles manually annotated as relevant	Recall	Number of articles retrieved via search string application	Precision
Spanish	2,113	104	0.92	105	0.93
English	3,418	170	0.88	165	0.90
German	1,203	119	0.89	111	0.94
Swedish	1,244	85	0.67	60	0.93
Polish	1,391	63	0.77	63	0.76
Hungarian	1,200	102	0.83	101	0.81
Romanian	1,415	63	0.71	61	0.71

Table 3 Boolean search strings used for retrieval of migration-related news articles

Country	Language	Search string
Spain	Spanish	asilo* OR inmigra* OR refugiad* OR migrante* OR migratori* OR "sin papeles" OR "campo de desplazados" OR patera* OR emigra* OR "libre circulación" OR "fuga de cerebros"
UK	English	asyl* OR immigrant* OR immigrat* OR migrant* OR migrat* OR refugee* OR foreigner* OR "undocumented worker*" OR "guest worker*" OR "foreign worker*" OR emigrat* OR "freedom of movement" OR "free movement"
Germany	German	asyl* OR immigrant* OR immigriert* OR immigrat* OR migrant* OR migrat* OR flüchtling* OR ausländer* OR zuwander* OR zugewander* OR einwander* OR eingewander* OR gastarbeiter* OR "ausländische arbeitnehmer*" OR emigr* OR auswander* OR ausgewander* OR personenfreizügigkeit* OR arbeitnehmerfreizügigkeit* OR "freier personenverkehr*"
Sweden	Swedish	asyl* OR invandr* OR migrat* OR migrant* OR flykting* OR utlänning* OR immigrant* OR ensamkommande* OR EU-migrant* OR "utländsk bakgrund" OR gästarbetar* OR "utländsk* arbet*" OR papperslös* OR emigr* OR utvandr* OR "fri rörlighet"
Poland	Polish	azyl* OR migr* OR imigr* OR uchodźca OR uchodźcy OR uchodźcę OR uchodźcą OR uchodźco OR uchodźców OR uchodźcom OR uchodźcami OR uchodźcach OR cudzoziem* OR obcokrajow* OR "robotni* z zagranicy" OR "pracowni* z zagranicy" OR gastarbeiter* OR "nielegaln* pracowni*" OR emigr* OR "swobodny przepływ"
Hungary	Hungarian	menedék* OR bevándor* OR immigrá* OR migrá* OR menekült* OR vendégmunk* OR elvándor* OR emmigrá* OR mozgásszabadság*
Romania	Romanian	azil* OR imigra* OR migra* OR emigra* OR refugiat* OR "muncitor străin" OR "muncitori străini" OR "muncitorii străni" OR "muncitorilor străni" OR "lucrător străin" OR "lucrători străini" OR "lucrătorii străini" OR "lucrătorilor străini" OR "libera circulație a persoanelor" OR "libertatea de circulație a persoanelor" OR "libera circulație a lucrătorilor" OR "libertatea de circulație a lucrătorilor"

Note: The search strings, and correspondingly the news articles, are in the most-widely spoken language for each country (e.g., not Catalan, Basque or Galician, but Spanish for Spain).

Article gathering and parsing

The strategy to gather articles was twofold, first articles were obtained from online news archives such as EMIS (see www.emis.com), second articles were scraped from online media websites. To do so, the final search strings (Table 3) were compiled to regular expressions and adapted to the requirements of the respective databases. We then used these to search for migration-related news articles and download them. Second, articles from online newspapers not available in the databases were scraped (including textual content and meta-data) using custom scrapers written in python. As websites offer a vast amount of articles in oftentimes multiple subsections, we decided to scrape the main (landing) page of every news website. To standardize this procedure, we furthermore scraped those pages once a day starting at 12pm. Only after articles were scraped and locally stored, the Boolean search strings (Table 3) were used to subset migration-related articles from the scrapped material.

Once all potentially relevant articles were gathered, unstructured information was extracted from the obtained material and organized in a data matrix, where each line represents an article. Using python for automation purposes, meta-information, such as publication date and source were extracted and added to the data matrix. Similarly, all textual contents (such as headline, lead paragraph, and main text) were identified and stored in separate columns of that matrix.

Data cleaning and preparation

Deduplication

A deduplication procedure was applied to the gathered data to eliminate duplicate articles that may arise due to faulty archiving, regional mutations of news outlets, or archiving of minimally-edited articles. To detect highly similar texts, we relied on the frequently used cosine similarity measure (Huang, 2008).

Comparing the textual content of two articles, the measure of cosine similarity indicates to what extent the articles resemble each other. Following a qualitative inspection of a sample of articles and their calculated cosine similarity, three of the authors determined country-specific thresholds for the identification of duplicates, thus accounting for language specificities (Spain: $\cos(\theta) = 0.98$; UK: $\cos(\theta) = 0.96$; Germany: $\cos(\theta) = 0.95$; Sweden: $\cos(\theta) = 0.95$; Poland: $\cos(\theta) = 0.95$; Hungary: $\cos(\theta) = 0.95$; Romania: $\cos(\theta) = 0.95$).

For each text, we checked for a duplicate ten days prior to its publication. Whenever a pair of articles exceeded this manually predefined and language-specific threshold, the shorter one of the two articles was excluded.

Stratified sampling and article translation

In a subsequent step, all non-English news articles were machine translated into English via the Google Translate API[2]. This was essential, as it allowed us to annotate the data with English language instruments for automated content analysis.

To save resources, only a sample of the initial texts was translated. A stratified sampling of every second article by outlet ensured that every outlet during the whole period of analysis would be represented in the data.

The final number of articles that are included in the provided data set are displayed in Table 4 (page 11–13).

[2] See https://cloud.google.com/translate/docs/

Table 4 (Part 1) Number of articles in the data set per media source

Country	Source type	Source	Number of Articles
Spain	Print	El Mundo,	916
		El Pais	1637
	Online	20minutos.es	440
		abc.es	573
		eldiario.es	1,004
		elmundo.es	425
		elpais.com	1,103
		larazon.es	547
		lavanguardia.com	650
UK	Print	Daily Mail	810
		Daily Mirror	634
		Metro	299
		The Daily Telegraph	1,246
		The Guardian	3,475
	Online	bbc.com/news/uk	215
		dailymail.co.uk	1,895
		express.co.uk	1,007
		mirror.co.uk	244
		theguardian.com/uk	1,465
		telegraph.co.uk	723
		thesun.co.uk	857
		thetimes.co.uk	981
Germany	Print	Bild	1,439
		Die Tageszeitung (taz)	2,038
		Die Welt	1,622
		Frankfurter Allgemeine Zeitung	2,826
		Frankfurter Rundschau	2,250
		Süddeutsche Zeitung	4,325

Table 4 (Part 2) Number of articles in the data set per media source

Country	Source type	Source	Number of Articles
Germany	Online	bild.de	810
		faz.net	1,420
		spiegel.de	901
		sueddeutsche.de	982
		taz.de	1,118
		welt.de	2,002
		zeit.de	1,368
Sweden	Print	Aftonbladet	672
		Expressen	67
		Svenska Dagbladet	1,094
	Online	aftonbladet.se	543
		dn.se	937
		expressen.se	830
		metro.se	201
		nyheter24.se	126
		svd.se	532
	Online / Alternative	etc.se	318
		friatider.se	543
		samnytt.se	673
Poland	Print	Dziennik Gazeta Prawna	337
		Gazeta Wyborcza	555
		Rzeczpospolita	897
Poland	Online	fakt.pl	185
		gazeta.pl	328
		onet.pl	414

Table 4 (Part 3) Number of articles in the data set per media source

Country	Source type	Source	Number of Articles
Poland	Online	rp.pl	918
		se.pl	161
		wp.pl	201
		wyborcza.pl	786
Hungary	Print	Magyar Hirlap	2,139
		Magyar Idök	2,139
		Nepszava	679
	Online	24.hu	752
		blikk.hu	152
		borsonline.hu	215
		index.hu	455
		magyarhirlap.hu	1,920
		mno.hu	536
		napi.hu	238
		nepszava.hu	844
		ripost.hu	784
Romania	Print	Evenimentul Zilei	78
		Jurnalul National	39
		Romania Libera	298
		Ziarul Financiar	146
	Online	adevarul.ro	330
		click.ro	99
		evz.ro	550
		jurnalul.ro	93
		libertatea.ro	390
		romanialibera.ro	54
		zf.ro	153
		ziare.com	460

Metadata

Most metadata were extracted automatically during the parsing of the raw text using python for automation purposes.

The meta information includes country of publication (`country`), the date of publication (`publication_date`), the time of publication (when available) (`publication_time`), the source (i.e., the media outlet that published the article) (`source`), and the source type (i.e., whether the media outlet is an online or print source) (`source_type`).

One meta information had to be computed separately, namely an outlets total coverage (`total_coverage`). Total coverage refers to each source's total number of articles on a given day including migration related but also non-migration related articles. This number was determined through automated access to the respective databases and search queries that did not employ any search terms or the total of all articles that appeared on the main page of news websites that we scraped on a daily basis. "Total coverage" thus refers to the total number of articles found within a database for a given source and a given day. In a final step, we estimated and subtracted the share of duplicate articles from the queried total coverage, based on the deduplication procedure applied to all migration-related articles (see section *Data cleaning and preparation* above).[3]

Based on the total coverage of a given source (s) in a given day (d), users may want to calculate the share of migration-related coverage in that given source and month.

$$Share\ of\ migration\ related\ articles_{sd} = \frac{migration\ related\ articles_{sd}}{total\ coverage_{sd}} * 100$$

However, please note that due to archive availability, there are some cases in the data set with missing information.

[3] i.e., if 15% of the migration-related articles in a specific month and source were duplicates, we subtracted 15% of articles from the total coverage in that month and source as well.

Key variables

In the following, we will define and describe the key variables of this media analysis as well as the tools that allowed us to measure them and annotate each article accordingly. All our tools make use of a so-called dictionary approach (Lind et al., 2019). Such an approach applies a top-down procedure, where texts are searched based on a predefined list of words and phrases that reflect the concept of interest. The rate at which specific keywords then appear (together) in a text are used to classify documents into substantive categories.

Intra-EU mobility

One key variable identifies whether an article refers to "Intra-EU mobility" (`european_mobil`). The annotated data set thus contains a variable that shows if an article mentions migration within the EU and/or Schengen area (coded with 1) or whether this is not the case (coded with 0).

Concept Definition

Here, "Intra-EU Mobility" is an umbrella term designed to capture all news stories referring to migration or people migrating (past/present/future) from a territory in the EU or Schengen area to another territory in the EU or Schengen area. Manual coders were provided with a list of EU or Schengen Area States[4], a definition of what was defined as migration (see page 7), and additional instructions such as:

> If people from territories that are not included in the list migrate, this is considered "Non-European Migration" and is thus not to be coded as "Intra-EU Mobility". It is important to keep in mind that migrants are not always referred to by mentioning their nationalities. Any references to a population group that is not country-specific but can certainly be attributed to intra-EU or Schengen migration is also to be coded as "Intra-EU mobility". For example, a) Northern/Central/Western Europeans/etc., b) EU-citizens/-people/-families/-nationals/etc., or c) people/families/nationals from the Eurozone or the Schengen area.
>
> Text examples for "Intra-EU Mobility" include:
>
> > "A Swedish person moves to Germany"
> > "Luisa from Poland moves to the UK for work"
> > "Northern Europeans often migrate to neighbouring countries",
> > "EU- citizens have the right to move within Europe"
> > "Ali recently received a German passport and can now study in France"

[4] 1. Austria, 2. Belgium, 3. Bulgaria, 4. Croatia, 5. Cyprus, 6. Czech Republic, 7. Denmark (also Greenland, Faroe Islands), 8. Estonia, 9. Finland, 10. France, 11. Germany, 12. Greece, 13. Hungary, 14. Iceland, 15. Ireland, 16. Italia, 17. Latvia, 18. Lichtenstein, 19. Lithuania, 20. Luxembourg, 21. Malta, 22. Netherlands, 23. Norway, 24. Poland, 25. Portugal, 26. Romania, 27. Slovakia, 28. Slovenia, 29. Spain (also Canaries), 30. Sweden, 31. Switzerland, 32. United Kingdom/UK (Northern Ireland, Scotland, Wales, England, Gibraltar)

The Dictionary

The automatic identification of "Intra-EU Mobility" coverage is made possible through an elaborate English-language search string (i.e. dictionary) that is applied to the lemmatized English version texts.

The dictionary uses a combination of different concepts. It includes the explicit mention of a country of origin or nationality as well as keywords referring to migration (e.g., emigration, mobility, free movement) or phrases like "EU mobility" or '"Erasmus student". Based on this list individual sentences within the corpora were identified and annotated as referring to "Intra-EU Mobility". If an article included at least one sentence referencing "Intra-EU Migration", it was annotated as "1"; otherwise as "0". It is important to note that a positive annotation does not exclude the possibility of other kinds of migration being also mentioned in an article.

The dictionary was validated manually. To establish intercoder reliability, three coders independently coded the same set of 100 randomly selected news articles, Krippendorff's alpha was $\alpha = 0.82$ (Krippendorff, 2013). Subsequently, $n = 840$ randomly selected news articles, were classified manually by the coders and automatically via the dictionary. Contrasting the manual and automated annotation decisions for $n = 840$ news articles, the final dictionary reached a satisfactory level of validity with recall = 0.69 and precision = 0.90. For details please see Table 5.

Table 5 Dictionary Performance (Recall and Precision) for the identification of Intra-EU Mobility coverage

Concept	Number of manually annotated articles	Number of articles manually annotated as relevant	Recall	Number of articles retrieved via dictionary application	Precision
Intra EU Mobility	840	98	.69	76	.90

Sentiment towards migration

To examine the sentiment of media coverage, we calculated three different sentiment values per article: (a) the sentiment of only migration-related sentences (`sentiment_mr_sentences`), (b) the sentiment of all sentences except migration-related ones (`sentiment_other_sentences`), and (c) the overall sentiment of all sentences in an article (`sentiment_overall`).

To quantify the sentiment, we used the pre-validated and frequently used Lexicoder sentiment dictionary by Young and Soroka (2012). The dictionary measures sentiment on the basis of 4,567 predefined positively and negatively connoted words, and quantifies the general valence of language.

To measure migration specific sentiment, within a specific article, we first selected all sentences containing migration-related words (see Table 3). We applied certain pre-

processing steps provided by Emily Luxon to improve the performance of the dictionary, involving the detection of negations or handling of punctuation.[5] All words within the selected sentences are then annotated based on the Lexicoder sentiment dictionary. Words that are not in the dictionary are assigned a neutral sentiment. Scores for each sentence are standardized according to the length of the sentence. Adding up all scores from words bearing positive sentiment (P_i), subtracting all scores from negative words (N_i), and dividing by the amount of words (W_i) in a sentence, we get a final score (S_i), revealing whether a sentence has a more positive or negative sentiment (see Kouloumpis, Wilson & Moore, 2011):

$$S_i = \frac{\sum P_i - \sum N_i}{\sum W_i}$$

A sentence's sentiment can theoretically range from -1 (all words within that sentence have a negative sentiment) to +1 (all words within that sentence have a positive sentiment), with zero referring either to a balance between positive and negative words or only neutral words within that sentence.

For the three sentiment scores provided in the data set, we…

> ... aggregated the sentiment of all migration-related sentences per article (`sentiment_mr_sentences`). This value thus reflects the general sentiment with which media tend to cover migration-related issues, events, or stories within an article.
>
> … we applied the same procedure to all non-migration-related sentences per article, giving the second sentiment score (`sentiment_other_sentences`). This value thus reflects the general sentiment with which media tend to cover non-migration-related issues, events or stories within an article.
>
> … we applied the same procedure to all sentences in an article, giving the third sentiment score (`sentiment_overall`). This value thus reflects the general sentiment within an article.

Migration-related frames

Furthermore, the annotated data set contains five variables that show whether an article contains aspects that relate to certain migration-related frames. Those variables are either coded as 1 if an article does contain references to the respective frame or as 0 if it does not.

Concept Definition

The economy, labour market, welfare, security, and culture frame have been identified to be among the most relevant frames in migration coverage (e.g., Eberl et al., 2018). Studying media coverage on migration with computer-assisted methods, we understand frames as topics formed through re-occurring patterns of specific words that help us to categorize

[5] See http://www.lexicoder.com/download/

documents (Jacobi, van Atteveldt, & Welbers, 2016). Table 6 gives a first overview about the operationalization and text examples per frame.

Table 6 Frame operationalization and examples

Frame	Operationalization (for validation)	Text example (article excerpt)
Economy	Does the article refer to economy/budget-related aspects of migration?	"Migration is an important part of being in a thriving, modern economy" (Source: The Guardian, machine translated)
Labour Market	Does the article refer to labour market-related aspects of migration?	"Refugee will provide information on the characteristic of job market in romania will be guide how to find a job and through battery of test of self - knowledge and self - assessment to find alternative in decision - making, development action plan and put it into practice" (Source: Romania Libera, machine translated)
Welfare	Does the article refer to welfare-related aspects of migration?	"The president of the PP does not see necessary to enumerate to the illegal immigrants so they can access basic public services, like health or education." (Source: ABC, machine translated)
Security	Does the article refer to security and/or crime-related aspects of migrants/migration?	"According to his father, the refugee Hussein K., who was accused in the Freiburg murder trial before the Youth Chamber, is considerably older." (Source: Die Zeit, machine translated)
Culture	Does the article refer to culture-related aspects of migrants/migration?	"Europe, he emphasised, had always had a multicultural identity and that required a culture of dialogue and the "respect [of] the foreigner, the immigrant and people from different cultures as worthy of being listened to". (Source: The Guardian)

The **economy-**related aspects of migration (`fr_eco`) include references to...

> … so-called "economic migrants", i.e., people referred to as being economically motivated to migrate
>
> … migration as being related to the economic development of the country of destination (e.g., individual migrants' hope for higher economic standard) or of the country of origin (e.g., emigration because of an economic crisis)
>
> … explicit economic (i.e., financial) consequences for the migrants' country of origin due to emigration (e.g., remittances)
>
> … explicit economic (i.e., financial) consequences for the migrants' country of transit or destination (host country) due to immigration (e.g., taxes)
>
> … any sort of costs for a country that arise from migration, e.g., direct payments for migrants, the handling of migration

All these aspects may be linked to the EU/country/county/municipal level.

The **labour market**-related aspects of migration (`fr_lab`) include references to...

> … migrants' participation in the labor market, this includes also actions/laws/plans to prevent/limit migrants' active participation in the labor market, and mentions of non-active/passive participation or unemployment
>
> … labour market-related consequences for the migrants' country of origin due to emigration
>
> … labour market-related consequences for the migrants' country of transit or destination (host country) due to immigration
>
> … legal and illegal forms of labour when related to migration
>
> … the "fear of foreigners" ONLY if it is clear from the context that the fear is related to labour market-related aspects (e.g., the fear of immigrants taking away jobs)
>
> … stories of individual migrants mentioning their working situation are also coded here (e.g., an artist with migration background, a soccer player playing abroad)

All these aspects may be linked to the EU/country/county/municipal level. An important specification is that politicians/professionals (e.g., social workers) dealing with aspects of migration are only part of the concept if the migration background of a politician/professional is explicitly mentioned. References to "economic migrant/migration" or "migrants contribute to the economy" are defined as being too implicit a reference to working/labour to be part of the labour market concept and are instead to be seen as part of the "Economy Frame".

The **welfare-**related aspects of migration (`fr_wel`) include references to...

> … welfare, public or social assistance, social benefits, social care, or social services
>
> … the following sub dimensions of welfare: (public) education, child and family support, work related support, pension/retirement, public healthcare, state subsidies food/clothing, public housing, and housing/accommodation organization for refugees/migrants
>
> … EU support/aid payments for countries to handle migration *if* the previously mentioned welfare aspects are specifically addressed, thus general mentions of "humanitarian aid", "european union aid" are not included

Welfare aspects to be included refer to services by countries of destination/transit or supranational institutions/organizations. Not included are monetary compensation/travel funding to leave a country. The mere mention of a reception center or refugee camp is not sufficient to code the "Welfare Frame" if it is only mentioned as mere place of events (e.g.,

a fire at a refugee camp, an attack on a public housing project or a meeting in a refugee accommodation); the article has to refer to handling/organization of accommodation of migrants (e.g., set up/create/manage/run a center/camp/accommodation).

The **security-**related aspects of migration (`fr_sec`) include references to...

> … security and crime issues when those held responsible are migrants
>
> … law and order, border security, border control, border protection actions (e.g., fence enforcement, tear gas), and also to the deportation of migrants
>
> … human trafficking/smuggling, subjective security feeling, police dealings with migrants, (fear of) terror connected to migration
>
> … the "fear of foreigners" ONLY if it is clear from the context that the fear is related to crime/security-related aspects (e.g., the fear of immigrants committing crimes)
>
> … illegal immigration

It is important that the aspects may be addressed not only against migration or migrants (e.g., migration threatens the security of our country) but also in a favourable context (e.g., illegal migration should be legalized, protest against the deportation of migrants, three refugees are deported despite their remarkable integration efforts in British society).

The **culture-**related aspects of migration (fr_cul) include references to…

> … European or national (collective) identity/tradition/customs/heritage/ancestry, values and beliefs discussed/evaluated in relation to migration/migrants
>
> … (collective) identity/tradition/customs/heritage/ancestry, values and beliefs of migrants as group (may or may not be in addition connected to religion, ethnicity)
>
> ... cultural rituals, practices and symbols, products and tools that are significant for group members, such as language, stories and literature (e.g., religious literature, mythologies, folklore) or music, when mentioned in reference to region, ethnicity, religion or shared identity based on common values/principles/practices; lifestyle-related cultural symbols including religious food or clothing, figures, events, locations, or objects
>
> … political and religious elements contributing to collective identity, values and beliefs.

For the purpose of this study, **culture** refers to a set of values, beliefs, principles and ideas of what is right vs. wrong, and subsequent rituals, practices and symbols shared by a specific group of individuals.

Dictionary creation and data annotation

The frame dictionaries for each of the five frames contain English keywords and were used to classify the English (translated) versions of the articles. The keywords are intended to be unambiguous in the English language and take into account both transnational discourses as well as country-specific sub discourses.

Given the high complexity and latent character of the concepts, the dictionary creation procedure was structured in several steps. Many of the keywords were initially selected from English language dictionaries that were identified in Lind and colleagues (2019). To uncover additional relevant keywords a sample of *n* = 2,000 randomly selected news articles from the text corpora where manually annotated by the researcher who created the dictionary. The 2000 articles included 296 Spanish, 290 English, 297 German, 283 Swedish, 266 Polish, 286 Hungarian, and 282 Romanian articles. The machine translated English versions of those were used for the following iterative refinement process:

- applying the latest draft version of a dictionary
- calculating recall and precision
- manually examining wrongly classified articles for keywords to be added, removed, or refined

These steps were repeated until recall and precision for all dictionaries were both above 0.67. Table 7 provides recall and precision values for the final version of the dictionaries.

Table 7 Dictionary performance (recall and precision) for the identification of frames in the refinement data set

Frame	Number of manually annotated articles	Number of articles manually annotated as relevant	Recall	Number of articles retrieved via dictionary application	Precision
Economy	2,000	434	.82	437	.81
Labour Market	2,000	585	.69	501	.81
Welfare	2,000	461	.69	427	.82
Security	2,000	848	.84	823	.86
Culture	2,000	665	.86	690	.83

Before annotation of articles, textual data has to be preprocessed accordingly. All words in the news articles are converted to lowercase and lemmatized via the udpipe R package (Wijffels, 2018). Each news article is segmented into sentences. Furthermore, keywords in the dictionaries are compiled as regular expressions in R.

For the annotation of articles with a “0” (= frame is not present) or “1” (= frame is present), frame-specific keywords that referred to the topics of economy, labour, welfare, security, or culture were only considered as qualifying to identify a migration-specific frame, if they appeared close to a migration-specific keyword in the text as defined above (see Table 3).

Relying on the R package stringr (Wickham, 2018), dictionaries then detect the number of matches between migration-related words and frame-dictionary keywords within a defined text unit. The definition of this text unit was a single sentence for the economy, labour market, welfare, and security frame, and up to three sentences for the culture frame. This was decided as for the economy, labour market, welfare, and security frame. Here finding both components within a single sentence was found to represent the best compromise between high precision and high recall and was thus used as basis for the annotation with a “1” (= frame is present). For the culture frame, a culture-specific keyword together with a migration-specific keyword within a text unit of up to three consecutive sentences showed a better performance - and was thus used as basis for the annotation with a “1” (= frame is present).

As results from single sentences or sets of sentences are aggregated to the level of each article, the minimum of one such frame-specific match (i.e., “frame is present”) is defined as sufficient to code an article as transporting the respective frame.

Validation procedure

The quality (i.e., validity) of the final version for the dictionaries was assessed based on a fresh subsample of *n* = 2,100 migration-related sentences. These sentences were randomly selected from the corpora and were not part of the articles used for the dictionary refinement. The sentences were manually annotated by a researcher who was not involved in the dictionary creation.

First, a shared understanding of the concepts between the researcher who created the dictionary (see previous step) and the researcher who manually coded this validation subset was assessed via an intercoder reliability test based on the coding of 100 migration-related sentences. The Krippendorff's alpha values (for Economy: .85, Labour Market: .93, Welfare: 1.00 Security: 1.00, Culture: .96) were deemed satisfactory. Table 8 shows recall and precision values based on the *n* = 2,100 articles for the five dictionaries.

Table 8 Dictionary performance (recall and precision) for the identification of frames in the validation data set

Frame	Number of manually annotated sentences	Number of sentences manually annotated as relevant	Recall	Number of sentences retrieved via dictionary application	Precision
Economy	2100	140	.82	130	.85
Labour Market	2100	192	.88	200	.84
Welfare	2100	134	.77	149	.87
Security	2100	339	.81	362	.86
Culture	2100	420	.85	390	.91

Migration-frame related sentiment

Having identified the presence of individual frames in news articles, this now allows us to automatically calculate five variables that represent the sentiment for each of these five frames.

To this end, first, all sentences[6] with frame-related keywords (e.g. for the economy frame a match between economy-related keywords and migration keywords) were identified. Second, each frame-related lemmatized sentence was annotated with the sentiment dictionary. Third, for each frame the resulting sentiment scores of the frame-related sentences were then aggregated (mean) on article level. This lead to five variables (`fr_eco_sent`, `fr_lab_sent`, `fr_wel_sent`, `fr_sec_sent`, `fr_cul_sent`).

Geographic regions

Oftentimes the geographic regions that are mentioned in relation to migration are of particular interest. In recent years the regions of (a) the Middle East (`middle_east`) and (b) Eastern Europe (`eastern_europe`) appear particularly relevant.

The annotated data set contains two variables containing information on the mentioning of these specific geographic regions in an article. “Mentioning these regions” means that the article contains a reference to at least one of the following...

> … a country in that region by name directly (e.g. “Bulgaria”)
>
> … the inhabitants of a country in that region, singular or plural (e.g. “Bulgarians”/”a Bulgarian”)

[6] for the culture frame, we here selected units of three sentences

… an adjective describing something or someone as being from/of origin from a country in that region (e.g. “Bulgarian”)

The dictionaries containing these three types of reference are in English language. The annotation was applied to the machine translated texts in the corpora. If any of the three types of reference were detected, the article was annotated as “mentioning the region” respectively. Accordingly, the variables are coded as “0” if a region is not mentioned and as “1” if it is mentioned.

Variable overview

Variable type	Variable description	Variable name
ID	unique article identifier	article_id
Metadata	country name	country
	article publication date (Format: YYYY-MM-DD)	publication_date
	article publication time (if available) (Format: HH-MM)	publication_time
	media outlet	source
	outlet type	source_type
	total coverage	total_coverage
Textual content	article headline, in original language	headline
Intra EU Mobility	migration from a territory in the EU or Schengen area to another territory in the EU or Schengen area	european_mobil
Sentiment	mean sentiment of all migration-related sentences	sentiment_mr_sentences
	mean sentiment of all non-migration-related sentences	sentiment_other_sentences
	mean sentiment of all sentences	sentiment_overall
Frames	economy frame	fr_eco
	labour market frame	fr_lab
	welfare frame	fr_wel
	security frame	fr_sec
	culture frame	fr_cul
Frame Sentiment	economy frame-related sentiment	fr_eco_sent
	labour market frame-related sentiment	fr_lab_sent
	welfare frame-related sentiment	fr_wel_sent
	security frame-related sentiment	fr_sec_sent
	culture frame-related sentiment	fr_cul_sent
Geographic Groups	reference to the Middle East	middle_east
	reference to Eastern Europe	eastern_europe

References

De Vreese, C. H., Boukes, M., Schuck, A., Vliegenthart, R., Bos, L., & Lelkes, Y. (2017). Linking survey and media content data: Opportunities, considerations, and pitfalls. *Communication Methods and Measures, 11*(4), 221-244.

Eberl, J.-M., Meltzer, C. E., Heidenreich, T., Herrero, B., Theorin, N., Lind, F., ... & Strömbäck, J. (2018). The European media discourse on immigration and its effects: A literature review. *Annals of the International Communication Association, 42*(3), 207–223.

Jacobi, C., van Atteveldt, W., & Welbers, K. (2016). Quantitative analysis of large amounts of journalistic texts using topic modelling. *Digital Journalism, 4*(1), 89–106.

Huang, A. (2008, April). Similarity measures for text document clustering. In *Proceedings of the sixth New Zealand computer science research student conference* (NZCSRSC2008), Christchurch, New Zealand (Vol. 4, pp. 9–56).

Kouloumpis, E., Wilson, T. & Moore, J. (2011, July). Twitter sentiment analysis: The good the bad and the omg!. In *Proceedings of the Fifth International AAAI conference on weblogs and social media*, Barcelona, Spain (The AAAI press, pp. 538–541).

Krippendorff K. (2013). *Content analysis. An introduction to its methodology.* Thousand Oaks, CA: Sage.

Lind, F., Heidenreich, T., Eberl, J.-M. & Boomgaarden, H. G. (2019). When the journey is as important as the goal: A roadmap to multilingual dictionary construction. *International Journal of Communication, 13*, 4000–4020.

Wickham, H. (2018). Stringr: Simple, consistent wrappers for common string operations (R Package Version 1.3.0) [Computer software]. http://stringr.tidyverse.org

Wijffels, J. (2018). Udpipe: Tokenization, parts of speech tagging, lemmatization and dependency parsing with the "UDPipe" "NLP" toolkit (R Package Version 0.6) [Computer software]. https://cran.r-project.org/web/packages/udpipe/index.html

Young, L. & Soroka, S., 2012. Lexicoder sentiment dictionary. [Computer software]. http://lexicoder.com